

Final Finite Element Assignment

Shear Wall Analysis

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1 Task presentation:

Figure 1 shows the graphical representation of the shear wall with holes, with necessary dimensions and other data.

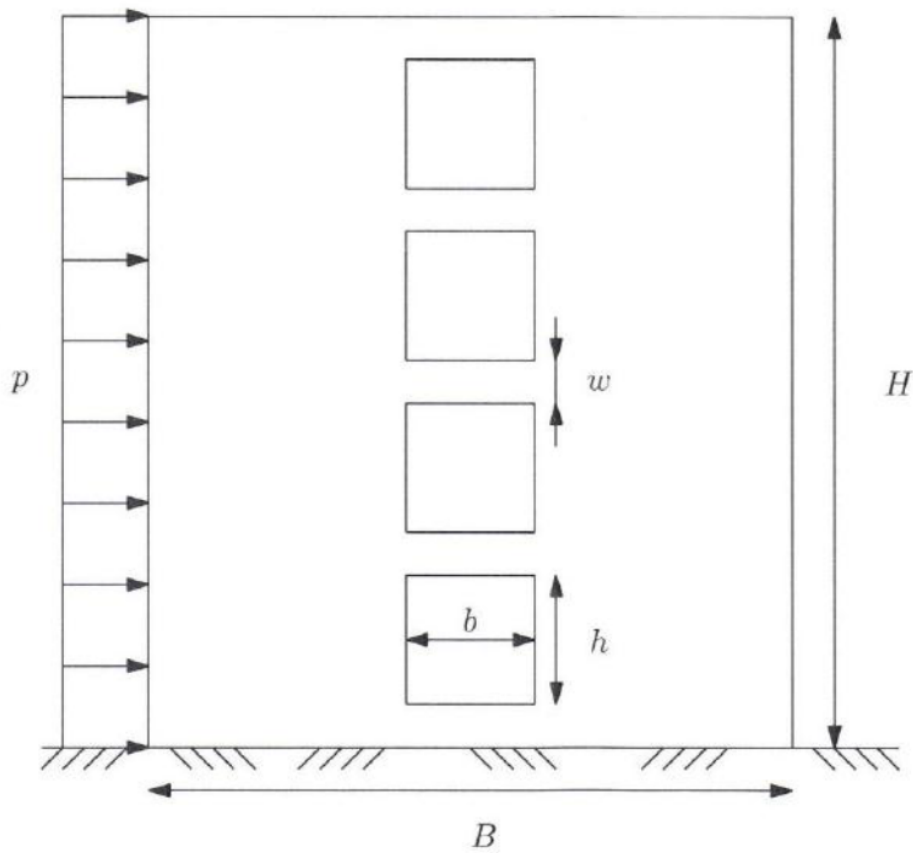


Figure 1: Shear wall geometry and dimensions

1.2 Boundary Condition:

$P = 25 \text{ kN/m}^2$ (Neumann boundary condition).

1.3 Dimension:

$$H = 14 \text{ m}$$

$$B = 14 \text{ m}$$

$$h = 1 \text{ m}$$

$$\begin{aligned}b &= 3 \text{ m} \\w &= 2 \text{ m} \\t &= 0.25 \text{ m}\end{aligned}$$

1.4 Materials Properties:

$$\begin{aligned}\text{Modulus of elasticity} & \quad E = 30 \times 10^9 \text{ Pa (N/m}^2\text{)} \\ \text{Mass density} & \quad \rho = 2400 \text{ kg/m}^3 \\ \text{Poisson ratio} & \quad \nu = 1/6\end{aligned}$$

2 Basic element:

The movement of the bottom in X and Y direction is restricted; moreover, in this problem the thickness, in comparison to other dimensions, has a small value, and we can assume we have a plane stress condition. This means that shear and normal stress components in Z direction are equal to 0.

2.1 Constitutive Equation

For isotropic materials the constitutive matrix $\mathbf{E}$ is:

$$\mathbf{E} = \frac{E}{1 - \nu^2} * \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1 - \nu}{2} \end{bmatrix}$$
$$\sigma = \mathbf{E} * \varepsilon$$

2.2 Compatibility Equation:

Our assumption in solving the problem is that during the deformation, the structure does not break, and no crack appears during stretching, and so on. This condition will happen if the following conditions are satisfied:

$$\varepsilon_x = \frac{\partial u}{\partial x} \quad \& \quad \varepsilon_y = \frac{\partial v}{\partial y} \quad \& \quad \gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$$

2.3 Equilibrium Condition:

For 2D condition, we can write:

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + F_x = 0$$
$$\frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + F_y = 0$$
$$[\partial]^T [\sigma] + [F] = 0$$

2.4 Boundary Condition:

We have two boundary conditions in this problem: Dirichlet boundary condition which refers to fixed boundary condition, and we can observe it at the bottom of the structure. On the left side of the structure we have Neumann boundary condition.

3 Analytical Solution:

A shear wall is a wall composed of braced panels, which is also known as shear panels, to counter the effects of lateral loads acting on a structure. We can match cantilever beam formulas with a shear wall through informed selection of a coordinate system (in this problem Y axis is vertical and X axis is horizontal). All the data which are mentioned below for the cantilever beam are extracted from *National Design Specification for Wood Construction* 2005 and other internet sources.

$$\begin{aligned}
 \text{force} \quad P &= -wx \\
 \text{structural moment} \quad M &= -\frac{1}{2}wx^2 \\
 \text{stress in X direction} \quad \sigma_M &= \frac{My}{I} \\
 \text{shear stress} \quad \tau &= \frac{PQ}{IT} \\
 \text{deflection at beam end} \quad \delta &= \frac{WL^4}{8EI} \\
 \text{deflection at point X} \quad \delta_x &= \frac{w}{24EI} (x^4 - 4l^3 + 3l^4) \\
 \text{angular rotation beam end} \quad \phi &= \frac{WL^3}{6EI} \\
 \text{moment of inertia of rectangular} \quad I &= \frac{ht^3}{12}
 \end{aligned}$$

$$p_0 = pt = 25000 * 0.25 = 6250 \text{ N/m}$$

Figure 2 is an example of a cantilever beam. This schematic figure indicates that the maximum bending moment happens in the fixed part (where there is Dirichlet boundary condition). That is why we should pay attention to this point (bottom of the shear wall). We also calculate the stress distribution in order to calculate deflection at the top of the shear wall.

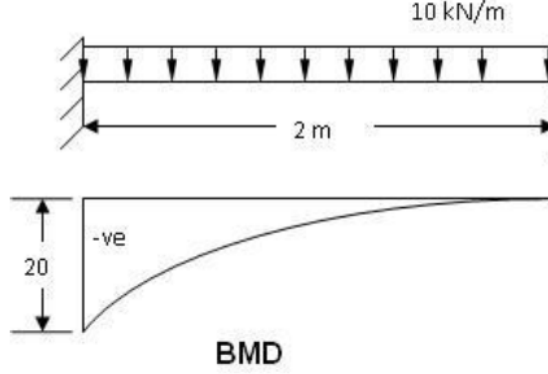


Figure 2: Cantilever-beam idealization for the analytical model

3.1 For $Y=0$

$$M = \frac{1}{2} * P_0 * (H - y)^2 = \frac{1}{2} * 6250 * 14^2 = -612500 \text{ Nm.}$$

$$I = \frac{B^3 t}{12} = \frac{14^3 * 0.25}{12} = 57.17 \text{ m}^4$$

In $Y = 0$ maximum normal stress will happen at the points 7 and -7 , in the condition that we place the neutral line $X = 0$ in the center of the shape.

For $X = 7$:

$$\sigma_{\text{Max } y, M} = \frac{Mx}{I} = \frac{-612500 * 7}{57.17} = -75000 \text{ Pa}$$

For $X = -7$:

$$\sigma_{\text{Max } y, M} = \frac{Mx}{I} = \frac{-612500 * (-7)}{57.17} = 75000 \text{ Pa}$$

The reason for different sign is that in bending, on one side we have tension while on the other side we have compression.

The weight of the shear wall should be calculated as a stress:

$$m = \rho * V = \rho * (BH - 4bh)tp = (14 * 14 - 4 * 3 * 1) * 0.25 * 2400 = 110400 \text{ kg}$$

where BH is the volume of shear wall in the absence of holes, and $4bh$ is the summation of hole volume.

$$\sigma_{y, \text{weight}} = \frac{-mg}{A} = \frac{110400 * 9.81}{14 * 0.25} = -309435 \text{ Pa}$$

then for $X = -7$:

$$\sigma_{y, \text{total}} = \sigma_{\text{Max } y, M} + \sigma_{y, \text{weight}} = 75000 - 309435 = -234435 \text{ Pa}$$

then for $X = +7$:

$$\sigma_{y, \text{total}} = \sigma_{\text{Max } y, M} + \sigma_{y, \text{weight}} = -75000 - 309435 = -384435 \text{ Pa}$$

for maximum shear stress:

$$Q = A'x' = \frac{Bt}{2} * \frac{B}{4} = \frac{14 * 0.25}{2} * \frac{14}{4} = 6.125 \text{ m}^3$$

$$P = P_0(H - y) = 6250 * 14 = 87500 \text{ N}$$

$$\tau_{xy, \text{maximum}} = \frac{PQ}{It} = \frac{87500 * 6.125}{57.17 * 0.25} = 37500 \text{ Pa}$$

This maximum shear stress happens in the center of the axes.

3.2 For Y=W

$$M = \frac{1}{2} * P_0 * (H - y)^2 = \frac{1}{2} * 6250 * 12^2 = -450000 \text{ Nm.}$$

$$I = \frac{B^3t}{12} = \frac{(14^3 - 3^3) * 0.25}{12} = 56.60 \text{ m}^4$$

Maximum normal stress will happen at the points 7 and -7.

For $X = 7$:

$$\sigma_{\text{Max } y, M} = \frac{Mx}{I} = \frac{-450000 * 7}{56.60} = -55654 \text{ Pa}$$

For $X = -7$:

$$\sigma_{\text{Max } y, M} = \frac{Mx}{I} = \frac{-450000 * (-7)}{56.60} = 55654 \text{ Pa}$$

The reason for opposite sign is mentioned before.

The weight of the shear wall which is located above the position $Y = W$ should be calculated as a stress:

$$m = \rho * V = \rho * (BH - 4bh)tp = (14 * 14 - (4 * 3 * 1) - (2 * 14)) * 0.25 * 2400 = 93600 \text{ kg}$$

$$\sigma_{y, \text{weight}} = \frac{-mg}{A} = \frac{-93600 * 9.81}{(14 * 0.25) - (3 * 0.25)} = -333897 \text{ Pa}$$

then for $X = -7$:

$$\sigma_{y, \text{total}} = \sigma_{\text{Max } y, M} + \sigma_{y, \text{weight}} = 55654 - 333897 = -278243 \text{ Pa}$$

then for $X = +7$:

$$\sigma_{y, \text{total}} = \sigma_{\text{Max } y, M} + \sigma_{y, \text{weight}} = -55654 - 333897 = -389551 \text{ Pa}$$

for maximum shear stress we should know that stress in 4 edges in $Y = W$ area is 0, with the highest shear stress between the two parts:

$$Q = A'x' = (7 - x) * t * \left(x - \frac{b}{2}\right) = 1.83 \text{ m}^3$$

$$P = P_0(H - y) = 6250 * 12 = 75000 \text{ N}$$

$$\tau_{xy, \text{maximum}} = \frac{PQ}{It} = \frac{75000 * 1.83}{56.60 * 0.25} = 9689 \text{ Pa}$$

3.3 Y=H

In $Y = H$ we have the highest value for deflection and rotation, which are calculated in this part:

$$I = \frac{B^3 t}{12} = \frac{14^3 * 0.25}{12} = 57.17 \text{ m}^4$$
$$\delta = -\frac{WL^4}{8EI} = -\frac{6250 * 14^4}{8 * 30 * 10^9 * 57.17} = -1.76 * 10^{-5}$$

for angular rotation at the top of the shear:

$$\phi = \frac{WL^3}{6EI} = -\frac{6250 * 14^3}{6 * 30 * 10^9 * 57.17} = -1.66 * 10^{-6}$$

In the above equation w is equal to q_0 .

4 Discussion FE model

4.1 Element type

Since in this problem thickness is very small comparing to height and width, load imposed on the model can be considered as a plane stress. As a result P183 has been chosen for the shear wall as the appropriate element for this problem and it is shown in Figure 3.

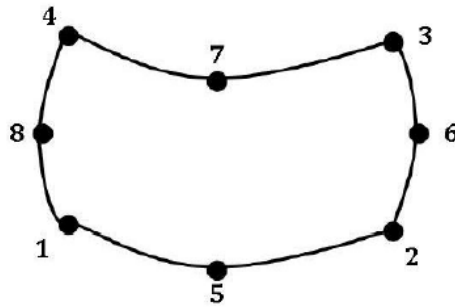


Figure 3: Schematic of P183 element

4.2 Element mesh

To investigate which mesh size is suitable for the problem, continuity of the nodal elements has to be taken into account. Since the problem does not have a complicated geometry and we just want to check the shear wall, it is not really necessary to choose a very fine mesh size, and as a consequence moderate mesh size of 0.25 has been chosen for this problem. The meshed model is illustrated in Figure 4.

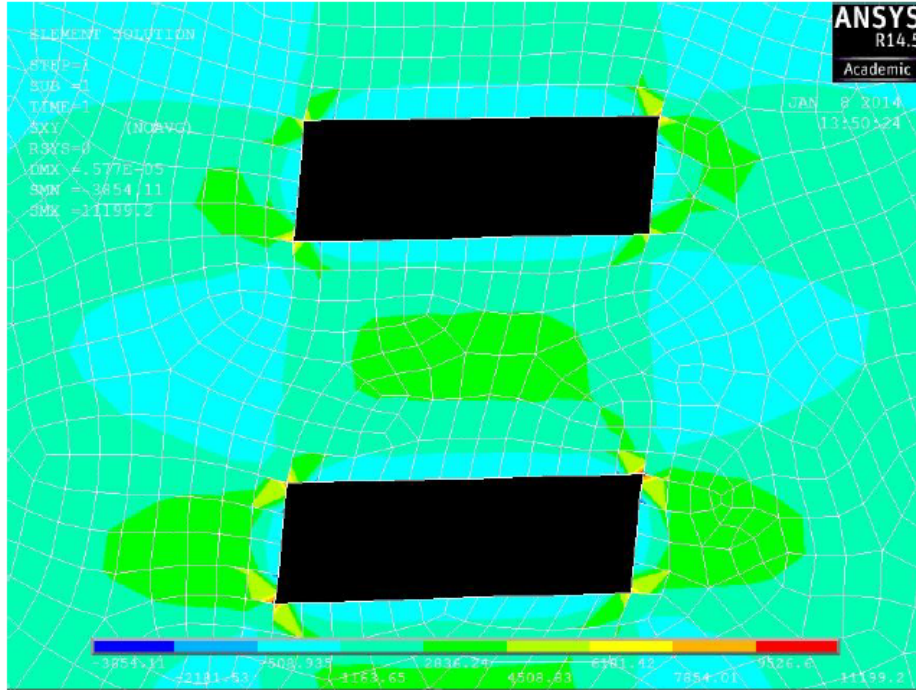


Figure 4: Mesh size of 0.25 with smaller discontinuity

4.3 Boundary conditions

For this problem there are two boundary conditions. Constant pressure is applied on the left side of the model, and the other one is that the nodal displacement of bottom side is stationary, which means that nodal displacement of the bottom side is equal to zero.

4.4 Symmetry

Pressure applied on the left side is not symmetric while shear stress is symmetric. The model can be divided in two parts: asymmetric and symmetric, but due to simplicity of the sample it is not necessary to do that, and as a result the whole sample is modeled.

5 Representation FE results

5.1 Displacement

Displacement of the meshed model is shown in Figure 5. Both deformed and undeformed shapes are shown in that figure. Also, it can be seen that the bottom side of the model is stable and stress is in the form of shear stress.

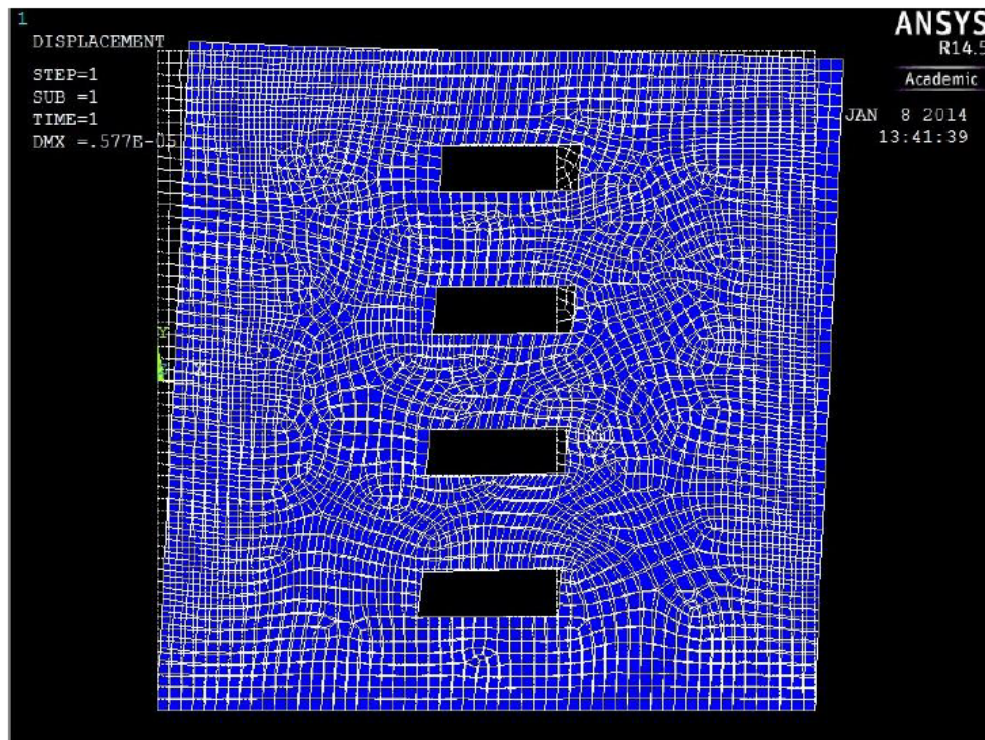


Figure 5: Deformed and undeformed shapes of the model

Nodal displacement in X direction is shown in Figure 6. It is shown that the maximum displacement is on the side that the load is applied and its amount equals to 0.577×10^{-5} m, which is on the top left. Besides, near the holes there are some fluctuations. It can be concluded that displacement is a bit bigger near holes than other areas in the model.

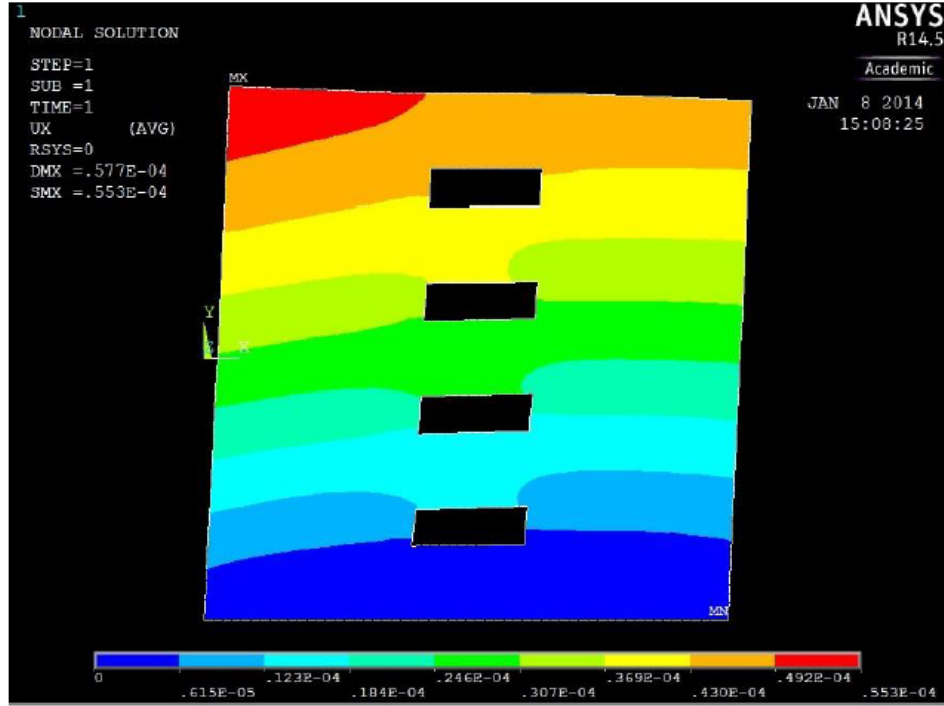


Figure 6: Nodal displacement in X-direction

Moreover, nodal displacement in Y direction is illustrated in Figure 7. The same as before, since the pressure is applied on the left side, maximum displacement is at the left side and it is equal to 0.163×10^{-5} . Furthermore, displacement at the bottom is almost equal to zero. At the right side of the model displacement is negative and it is normal in models with shear stress.

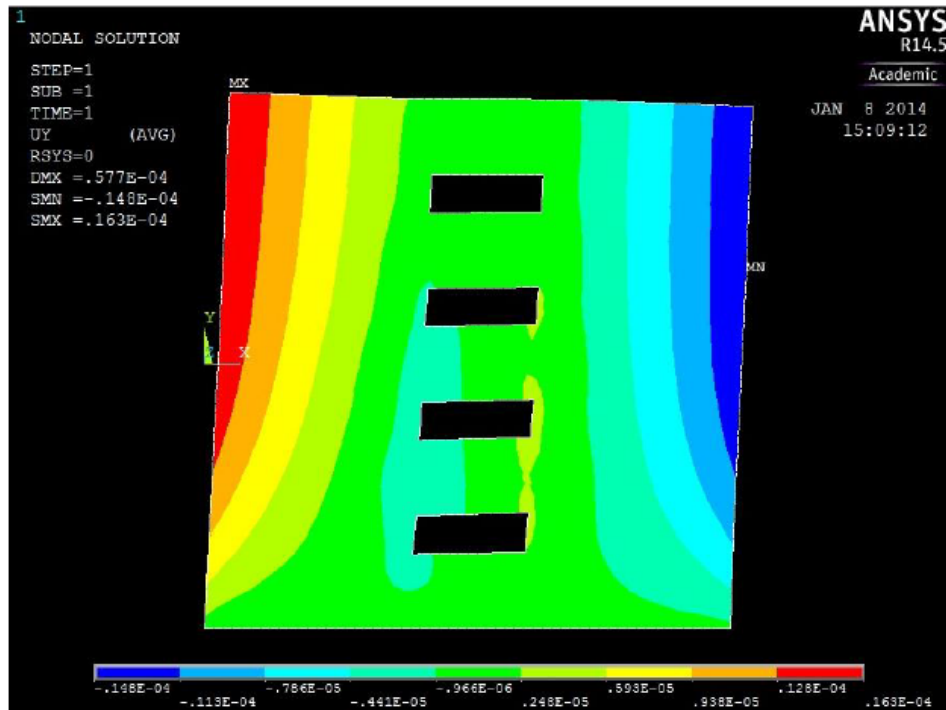


Figure 7: Nodal displacement in Y-direction

Figure 8 shows the sum of the nodal displacement. Maximum displacement happens at the top left with the amount of 0.577×10^{-5} m.

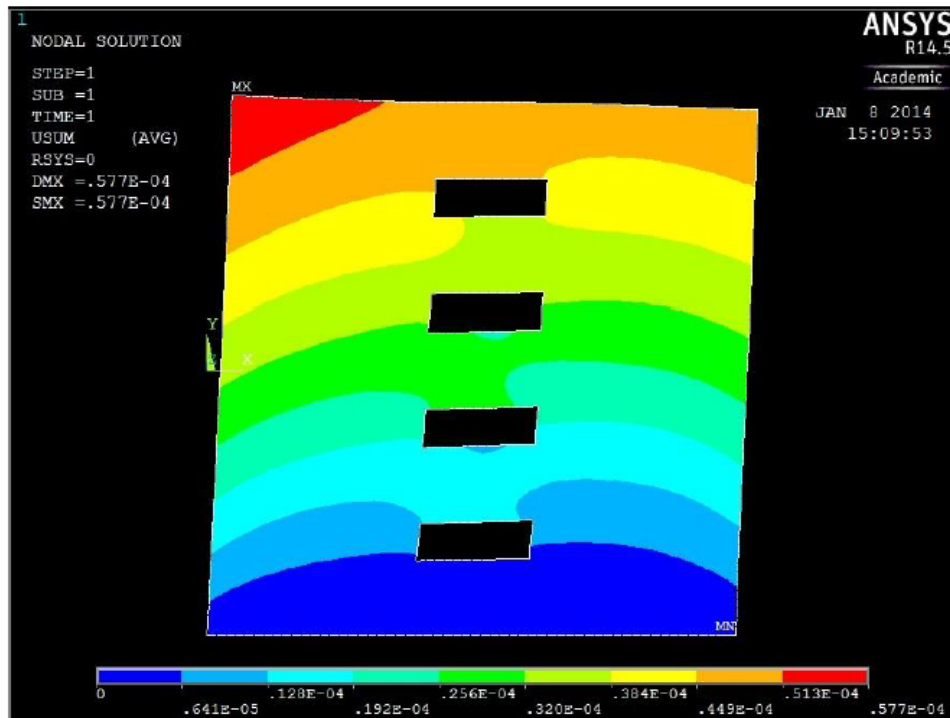


Figure 8: Sum of the nodal displacement

5.2 Stress

Two kinds of stress can be defined: nodal stress and element stress. Actually the same results of stress are seen in different ways. Nodal stress is more suitable for having a general idea about the stress in the model. Element stress checks if meshes are necessary to be refined.

5.2.1 Nodal stress

Figure 9 shows the nodal stress in X-direction. It is obvious that this kind of stress is maximum near holes. Red spots show the stress of nodes on the top left and bottom right of the holes and the maximum amount is equal to 125480 Pa, and for the top right and bottom left it is equal to 162445 Pa.

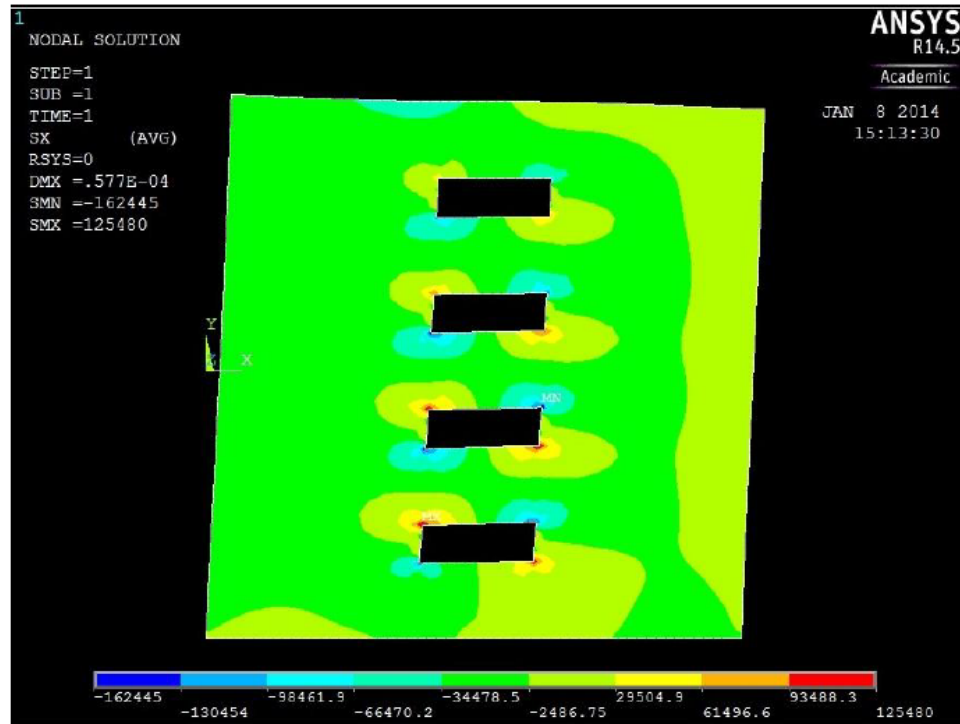


Figure 9: Nodal stress in X-direction

Nodal stress in Y direction is shown in Figure 10. Bottom corners have the most nodal stresses; left corner stress is tension and the right one compression, which are equal to 188043 Pa and 304715 Pa respectively. This behavior can be explained due to the stable bottom of the model.

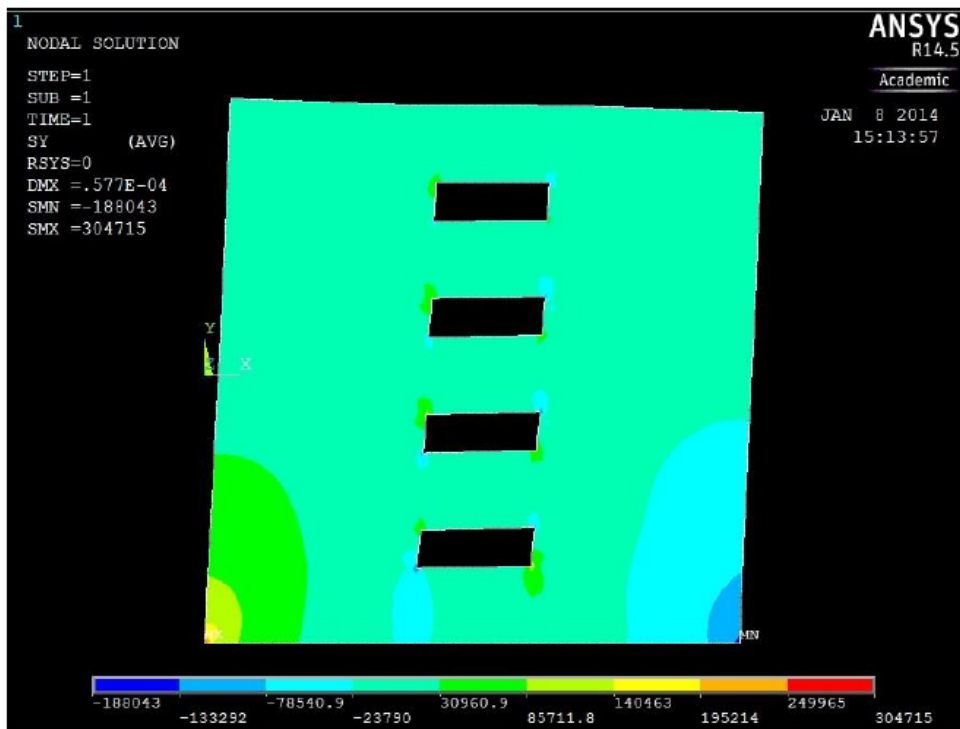


Figure 10: Nodal stress in Y-direction

Since we assumed that the stress is in a 2-dimensional plane stress, stress in Z direction is equal to zero, and it is illustrated in Figure 11.

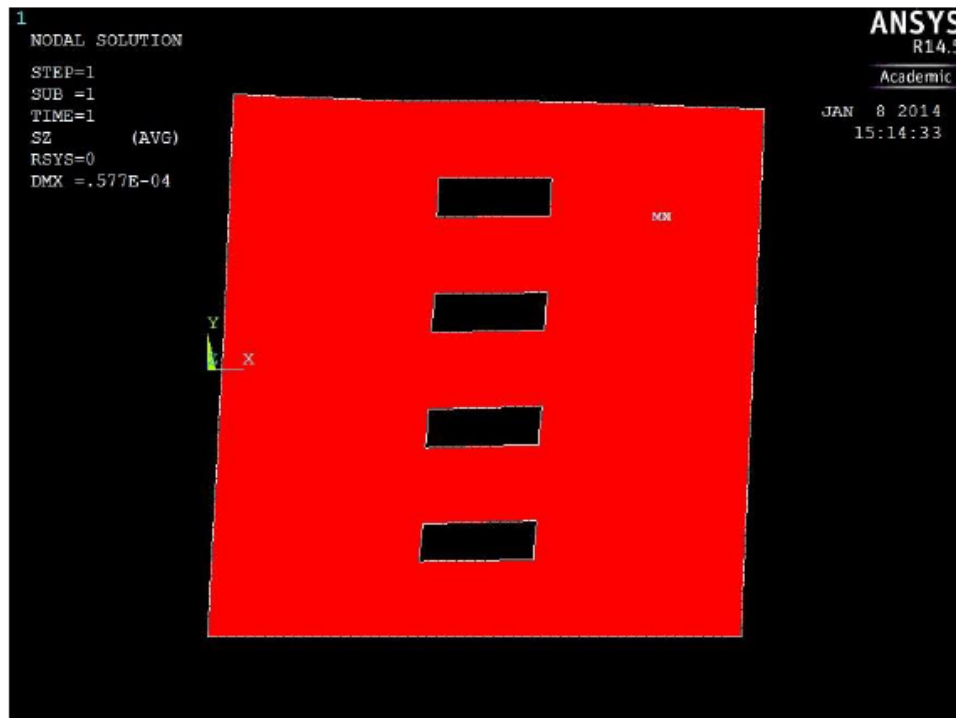


Figure 11: Nodal stress in Z-direction

The nodal stresses of the model in the x-y plane are plotted in Figure 12. It can be seen that shear stress is maximum at the bottom left and as it is mentioned before it is because of the stable bottom side (zero nodal displacement).

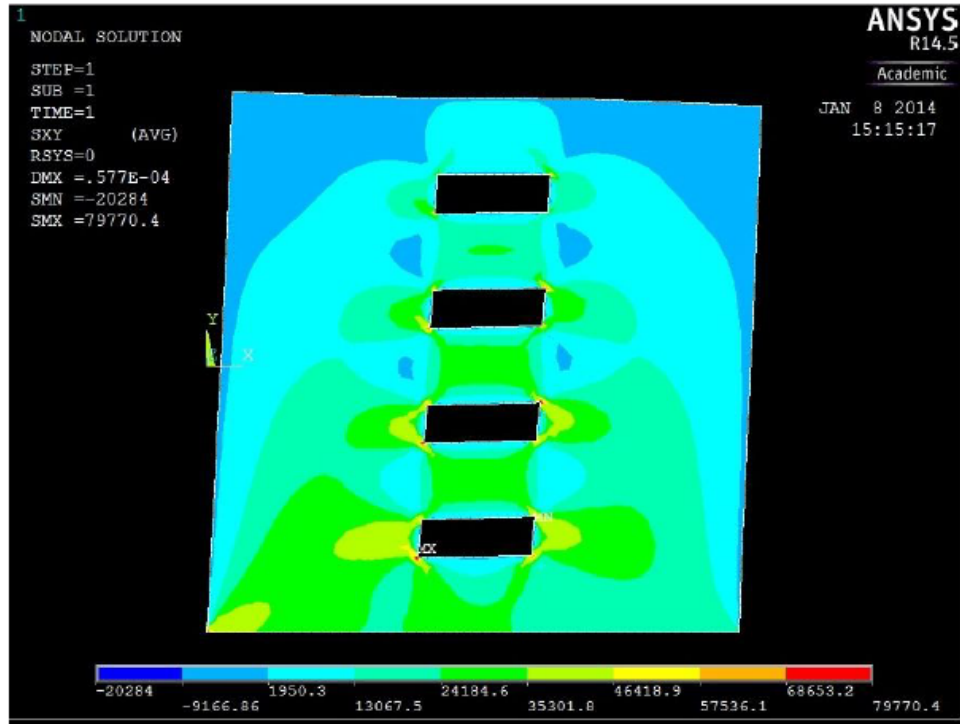


Figure 12: Nodal shear stress in XY-plane

5.2.2 Element stress

Element stresses are shown in X, Y, and XY direction like the previous section. Element stress in X direction is shown in Figure 13. Element stress is almost the same as nodal stress. Corners at top left and bottom right of the holes in the model have the maximum tension stress value of 190502 Pa while the other two corners have the maximum compression value of 242651 Pa. Like before, left side of the model has compression stress and right side has tension stress. It can be concluded that this model shows the necessity of mesh refinement.

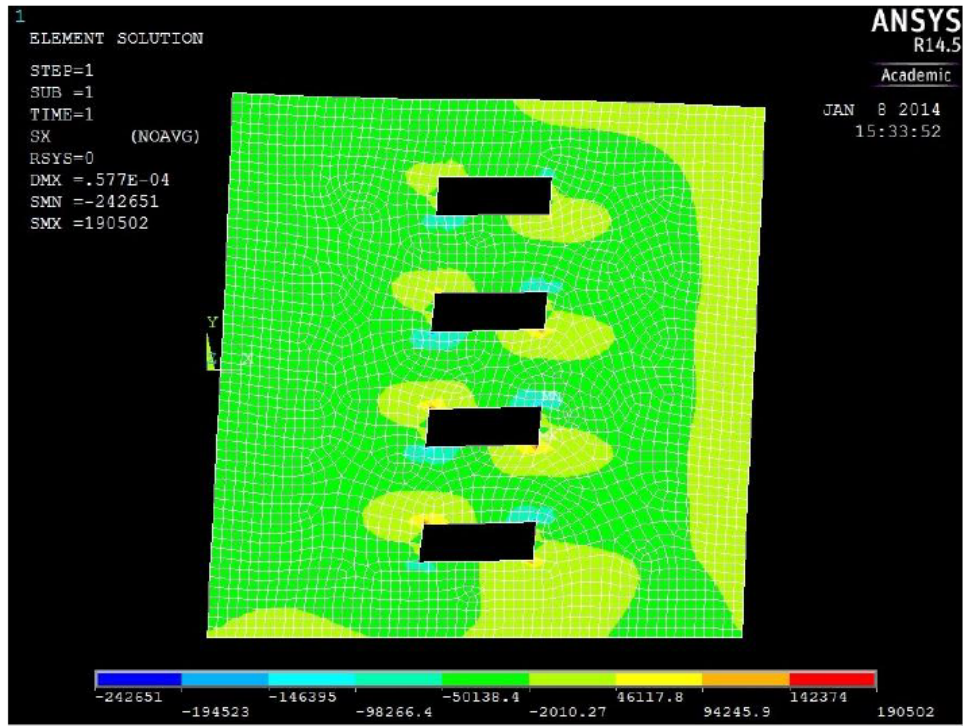


Figure 13: Element stress in X-direction

Element stresses in other directions are shown in the following Figure 14 and Figure 15. The general distribution is almost the same as the nodal stress in those directions, but values are different.

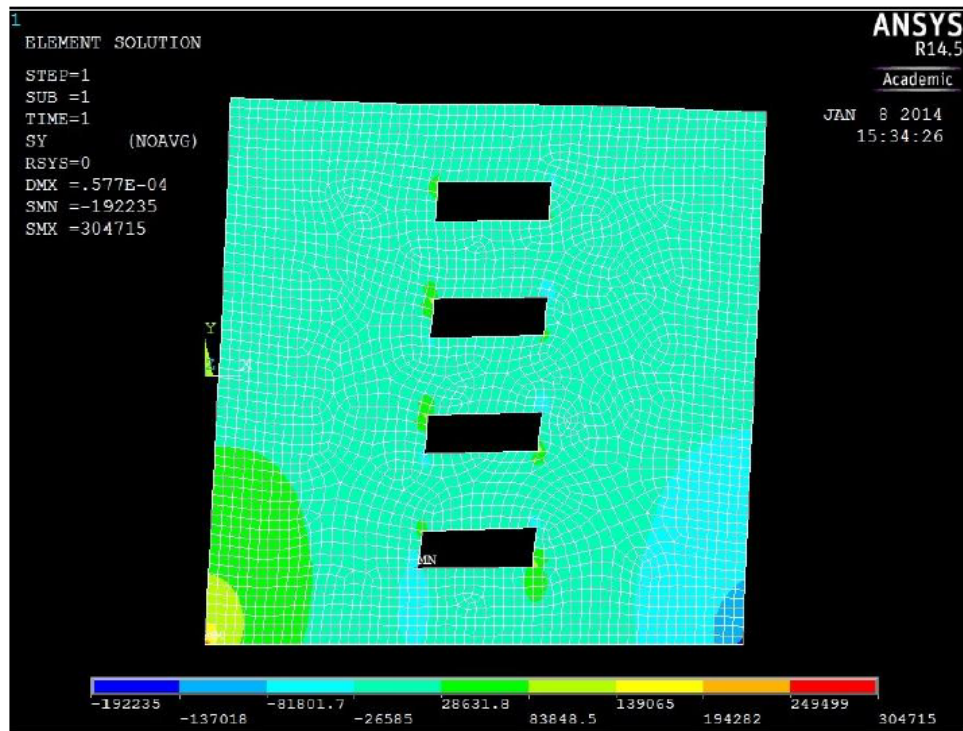


Figure 14: Element stress in Y-direction

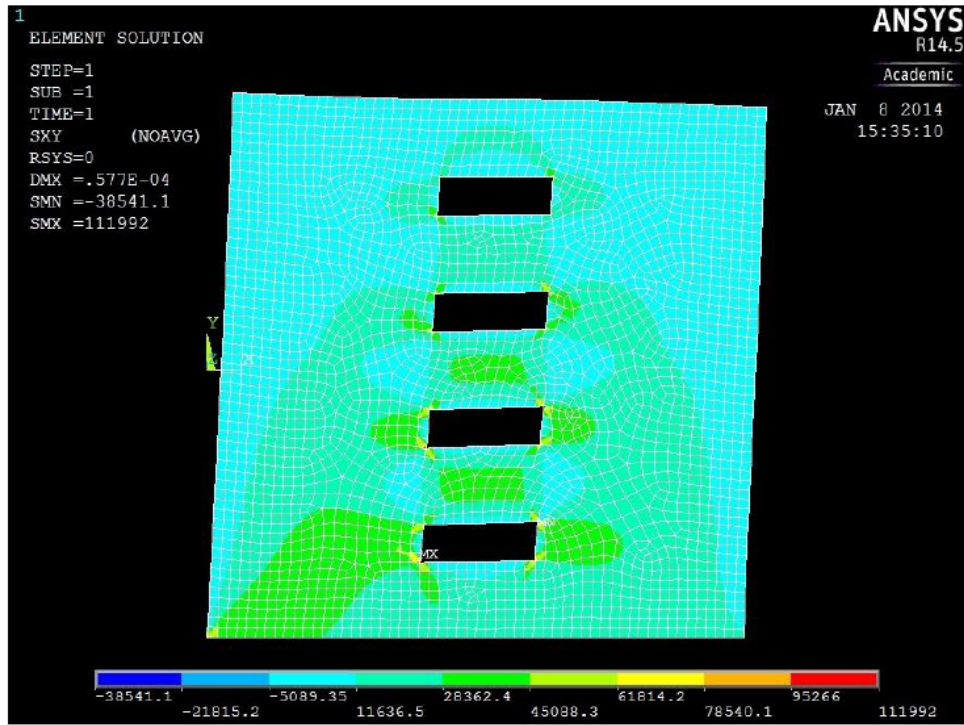


Figure 15: Element shear stress in XY-plane

5.3 Force

Figure 16 shows the structural force in X direction. At the bottom left it can be seen that there are some forces that are the result of bottom side with zero displacement. Moreover, around the holes there are some forces with values of 3460.49 N and 3215.96 N.

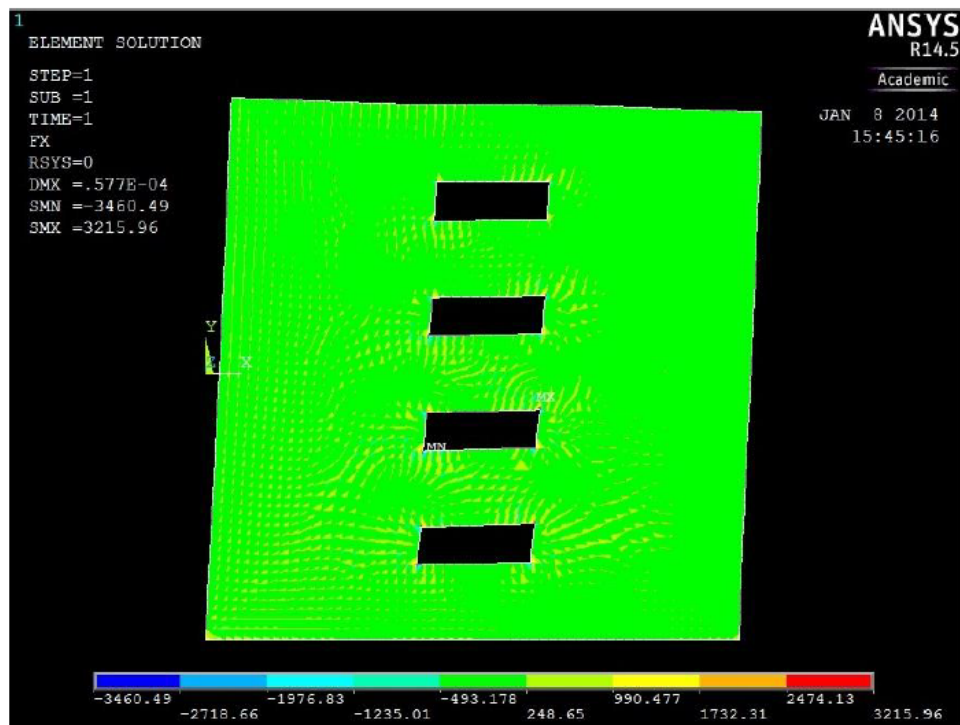


Figure 16: Forces in X-direction

6 Comparison of FE model and Analytical model

Equilibrium equations, stress-strain relation and strain-displacement relations have to be investigated in this part. By calculating stress and shear stress in different directions in the model, results are shown in Table 1. The difference between ANSYS and analytical values is illustrated. It is shown that there is a significant difference between ANSYS and analytical analysis of the model especially in Y-direction. It can be explained by neglecting the effect of gravity in the analytical model, which is considered in ANSYS model.

Table 1: Difference between ANSYS and analytical value

	Analytical	ANSYS	DIFFERENCE
σ_y	-287456	-271232	6%
σ_y	-393456	-184322	53%
τ_{xy}	9783	24876	60%
τ_{xy}	38493	29343	23%

7 Control calculations

7.1 Nodal equilibrium

Elements and nodes are shown in Figure 17 with different colors and numbers. Due to Cauchy law, sum of shear stress and normal stresses defines the acceleration of the body in one direction. Since our model is rigid, sum of all the forces at one node should be equal to zero. To prove that, some mathematical operation is done:

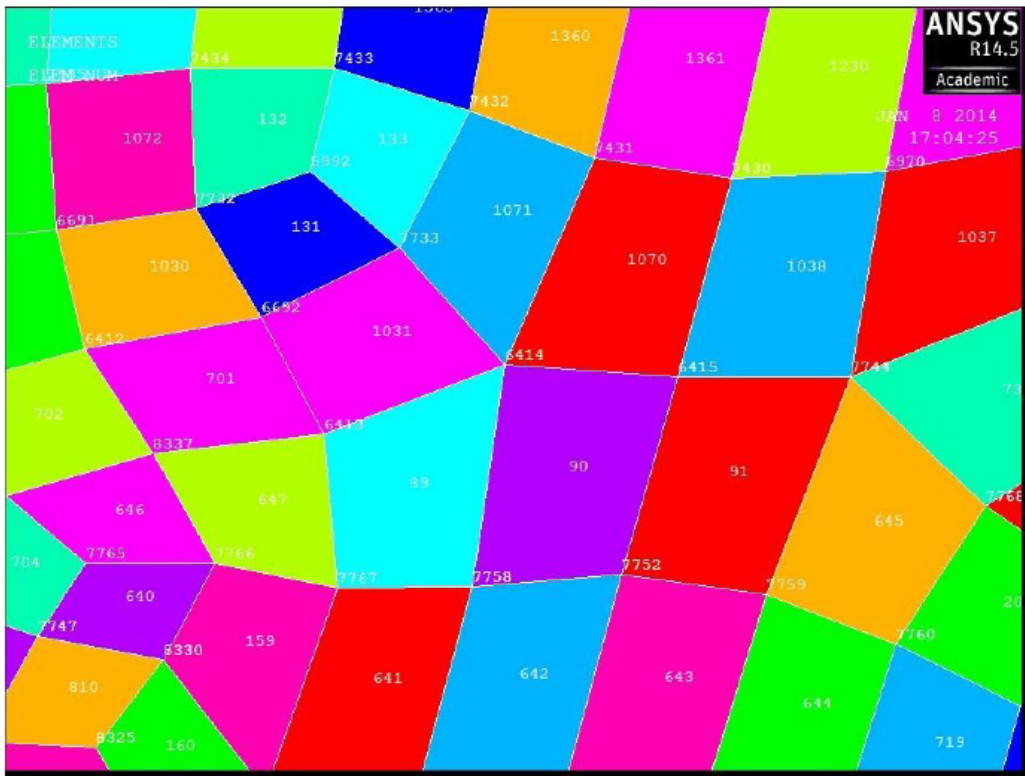


Figure 17: Nodes and Elements

Table 2: Overview of nodal forces

Node	Element	Force X	Force Y
6415			
	1070	391.32	433.32
	1038	-111.53	342.5
	90	-424.65	-643.54
	91	144.852	-475.069
Sum		0.08	0.011

And as a result it can be seen that this sum is almost equal to zero.

7.2 Element equilibrium

Elemental equilibrium is also checked by summing all the nodes of one element to see if it equals to zero or not.

Table 3: Overview of element forces

Element	Node	Force X	Force Y
91	7752	-151.65	450.04
	7759	834.10	-417.22
	7744	67.435	-486.63
	6415	-772.20	468.84
	4397	1033.9	130.37
	5534	2101.9	-1917.2
	4363	-1547.6	42.794
	782	-1565.9	1729.0
Sum		-0.15	-0.006

7.3 External equilibrium

For checking the general equilibrium of the model, forces applied on the model should be compensated by the reaction forces. To prove that, applied force on the ground should be equal to the integral of shear forces and the vertical forces. Figure 18 shows:

$$F_H = \int_0^{14} \sigma_x dx.$$

At the end of the model ($x = 14$) this amount is equal to 350000 and it is equal to 87500 N. It can be seen it is the same as force in horizontal direction. Moreover, almost the same result has been obtained for moment.

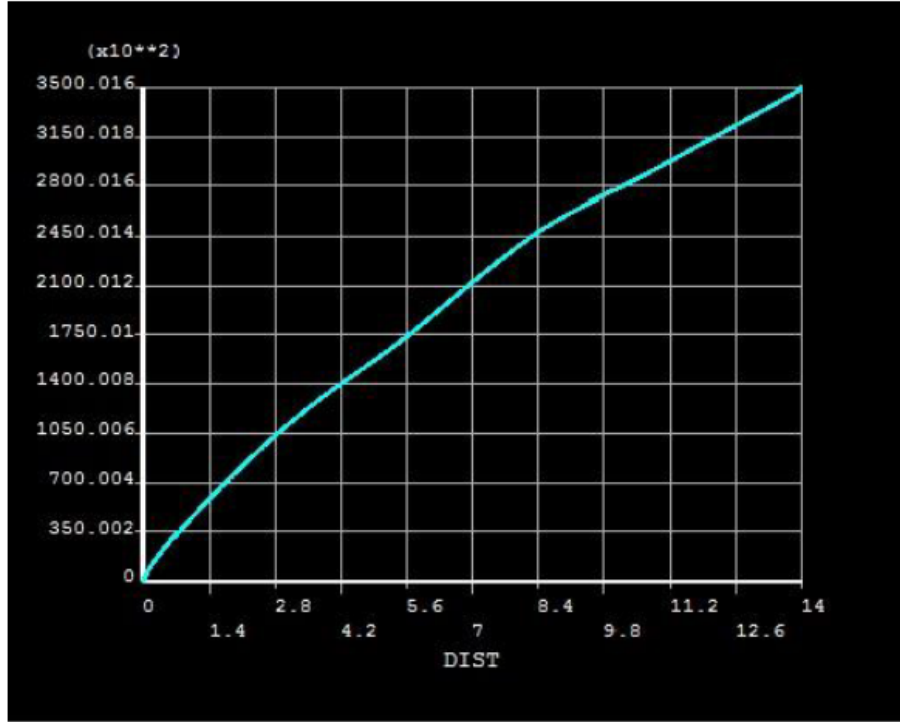


Figure 18: Shear force integral as a function of X

7.4 Stress Continuity

Continuity of the stress in finite element model between elements is an important factor since it shows whether this simulation is good or not. To investigate this factor, some nodes are taken for related calculation.

Table 4: Mean value of S_x, S_y, S_{xy}

Node	Elements	S_x	S_y	S_{xy}
6414	1070	-3427.8	-20486	23685
	1071	-3456.5	-20475	23701
	89	-3436.6	-20482	23694
	90	-3435.3	-20479	23689
	1031	-3428	-20468	23693
Mean		-3436.84	-20478	23692.4

Then considering the confidence (95%) we will have some errors which are shown in the following Table 5:

Table 5: Errors calculated for S_x, S_y, S_{xy}

	S_x	S_y	S_{xy}
Confidence (95%)	17.9791	11.9891	14.653
Error	0.523	0.586	0.613

It can be seen that the error is large. As a consequence, a finer mesh is needed for this model.

8 Evaluation of the element mesh

Several investigations have been done on the model. It is obvious that there are differences between FE model and analytical model. As it is mentioned before, these differences can be reduced by a finer mesh or using more complex mathematical functions. Moreover, continuity of the model and elements has been checked. More investigation shows that corners of the holes do not match with the mesh size. Furthermore, there is an error for this model which is equal to 12.452 according to the error estimation.